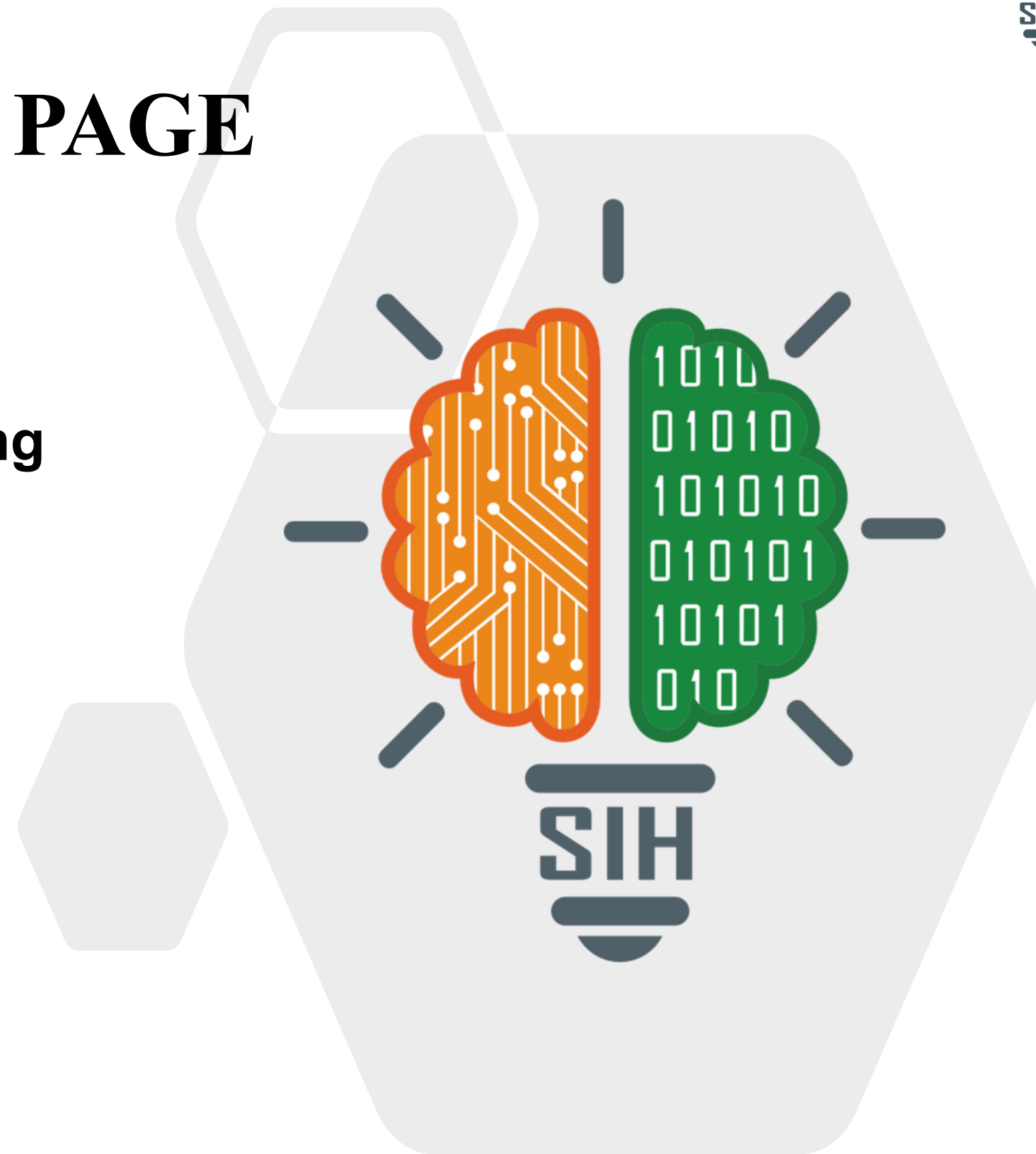


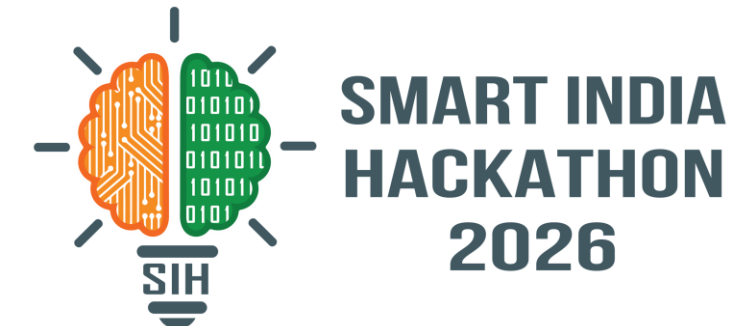
TITLE PAGE

- Problem Statement ID - SIH26066
- Problem Statement Title - OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations
- Theme - Disaster Management
- PS Category - Software
- Team ID - K26125
- Team Name - Team Null



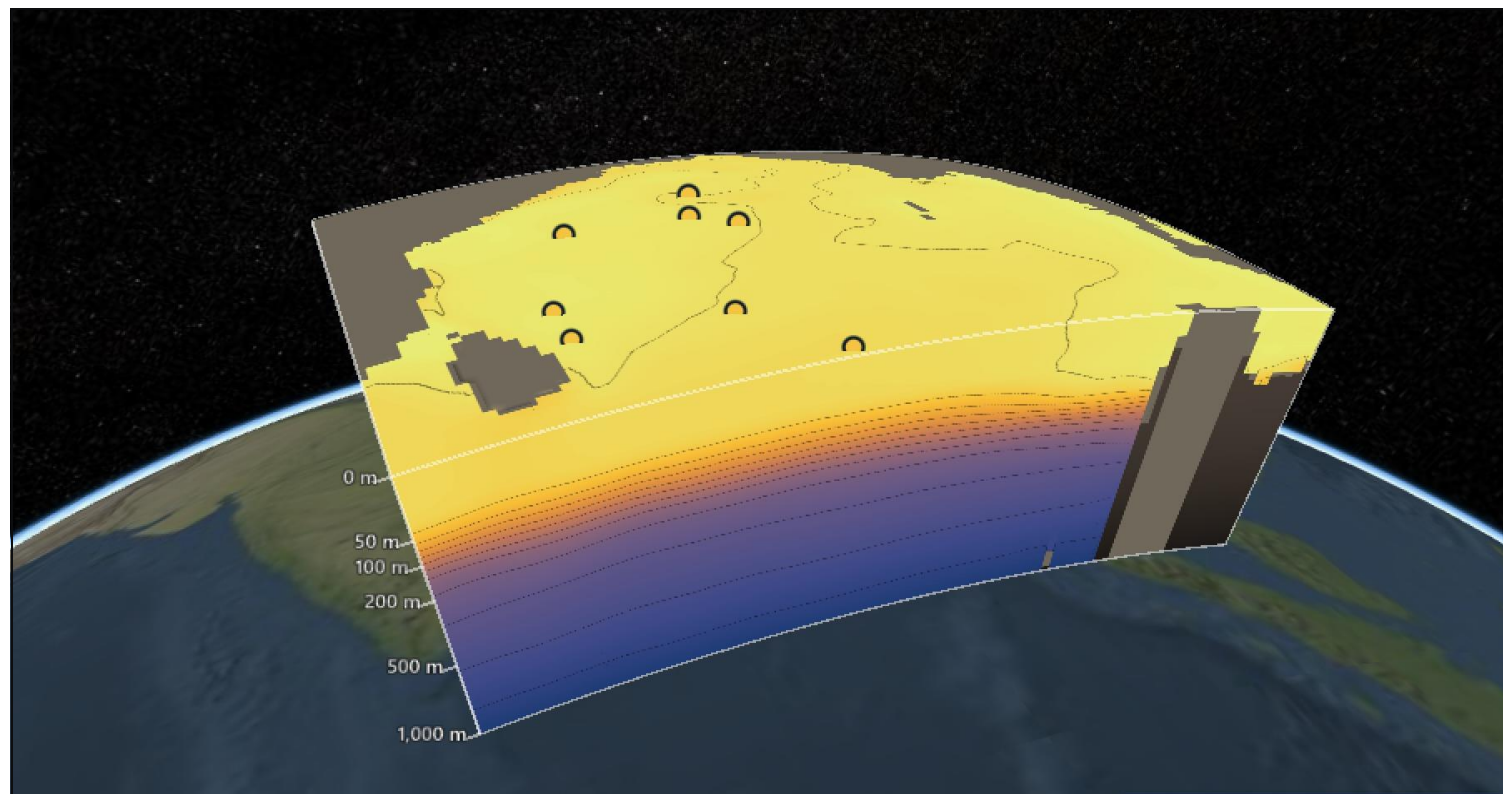


Satelight: Ocean from Space



PROPOSED SOLUTION

Eight satellite surface fields in, **temperature at 15 depths to 1,000 m out, every day at 0.25°**, over the North Indian Ocean, through a learned embedding.



Working prototype (idea stage), captured. The Bay of Bengal on 2023-05-17, after Cyclone Mocha, reconstructed from satellites alone to 1,000 m. Bottom right: held-out Argo float 2902766 cycle 122, data mode D, 0 levels failed QC, file D2902766_122.nc.

DETAILED EXPLANATION

Inputs: SST, salinity, sea level (SSH, SLA), currents and winds, harmonised onto one 0.25° daily grid

Model: CNN + transformer encoder makes the embedding; a decoder turns it into 15 levels; trained on GLORYS12, 2010–2020

Proof: 6,144 held-out Argo casts, 2023–24, scored per depth and basin beside climatology and GLORYS

HOW IT ADDRESSES THE PROBLEM

The expected solution, point by point, and what is built.

Preprocessing and harmonisation of multi-source data
one fetch per source, block means onto 0.25° daily; every assumption travels in the file; a re-fetch is bit-identical

A satellite embedding engine
each cell's surface state compressed to 32 numbers, saved per day as a file you can open, cluster and probe

Deep-learning reconstruction at 15 standard depths
exactly 0 to 1,000 m on the asked levels; cells shallower than a level are masked, never extrapolated

Validation with independent Argo
RMSE, bias and r per depth and basin; 8,500 levels failing QC counted, never dropped

Working PoC over the Bay of Bengal and Arabian Sea
a 3D viewer: the inputs, the embedding, the reconstruction beside GLORYS, and the held-out floats, day by day

Also asked for by name, and built

Daily NetCDF output, 0.25° · INCOIS gridded Argo as a second reference · a run without currents and winds · every run from a config and a seed

INNOVATION AND UNIQUENESS

Scored between a floor and a ceiling
every number beside climatology and GLORYS; at the surface it closes 75% of the gap between them

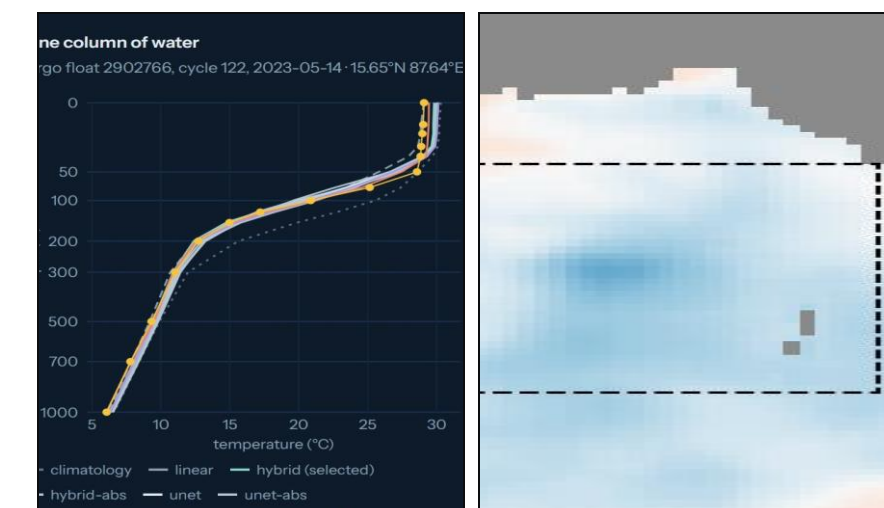
What a cyclone feeds on
20 °C and 26 °C isotherm depths and heat potential, each checked against Argo

Sees a cyclone's cold wake
Mocha and Biparjoy, 2023, reconstructed from satellites only

The reference is checked too
INCOIS gridded Argo range-tested and scored against Argo itself, before anyone is scored against it

No leakage, by design
contiguous time blocks; GLORYS never an input; the test years never touched in training

An embedding you can open
saved per day, clustered and probed; where it falls short, the deck says so



a float's profile vs every model

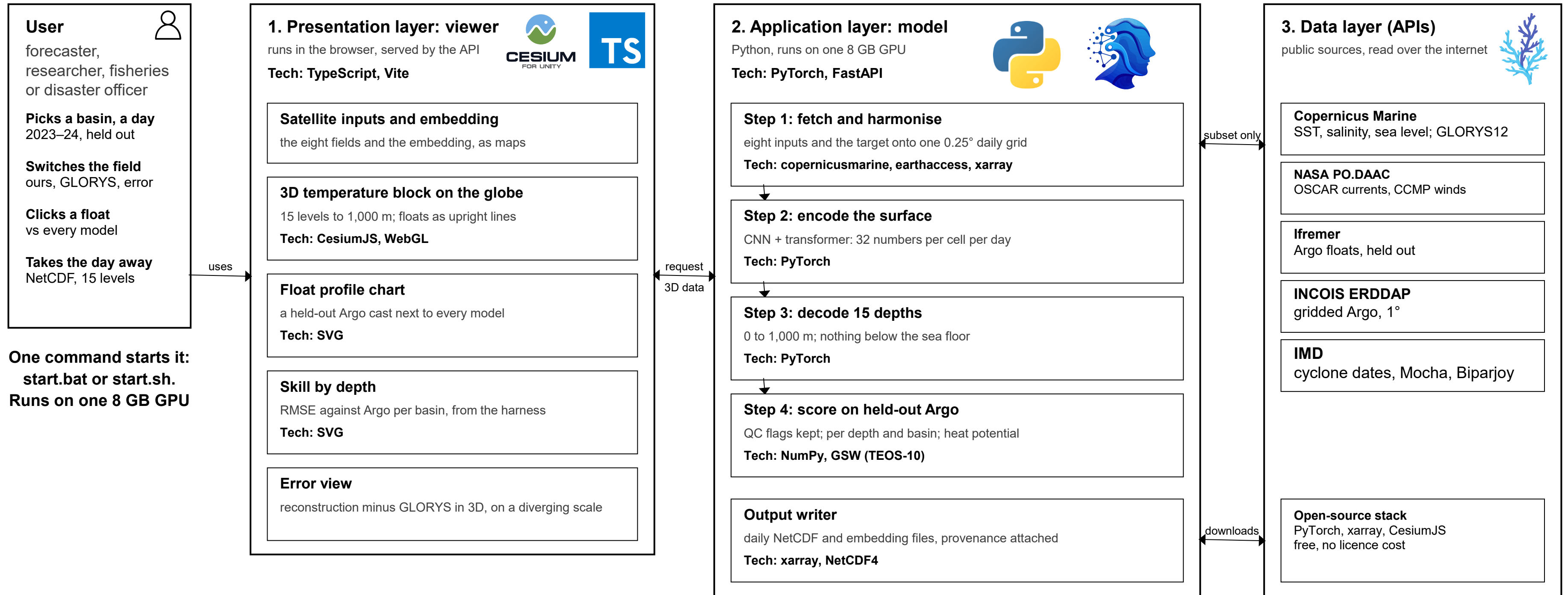
Mocha's cold wake, satellites only



TECHNICAL APPROACH



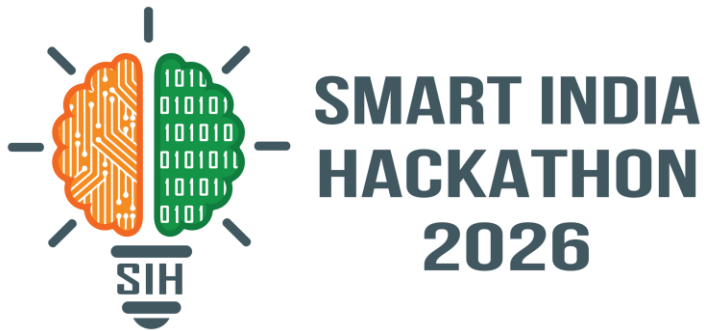
Software architecture: three layers, end to end



What the model writes for each cell and day: temperature at 0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700 and 1,000 m



FEASIBILITY AND VIABILITY



ANALYSIS OF FEASIBILITY: MEASURED BY THE EVAL HARNESS

6,144

held-out Argo casts scored, 2023–24; the model never saw one of them

711

test days reconstructed and scored, one file per day

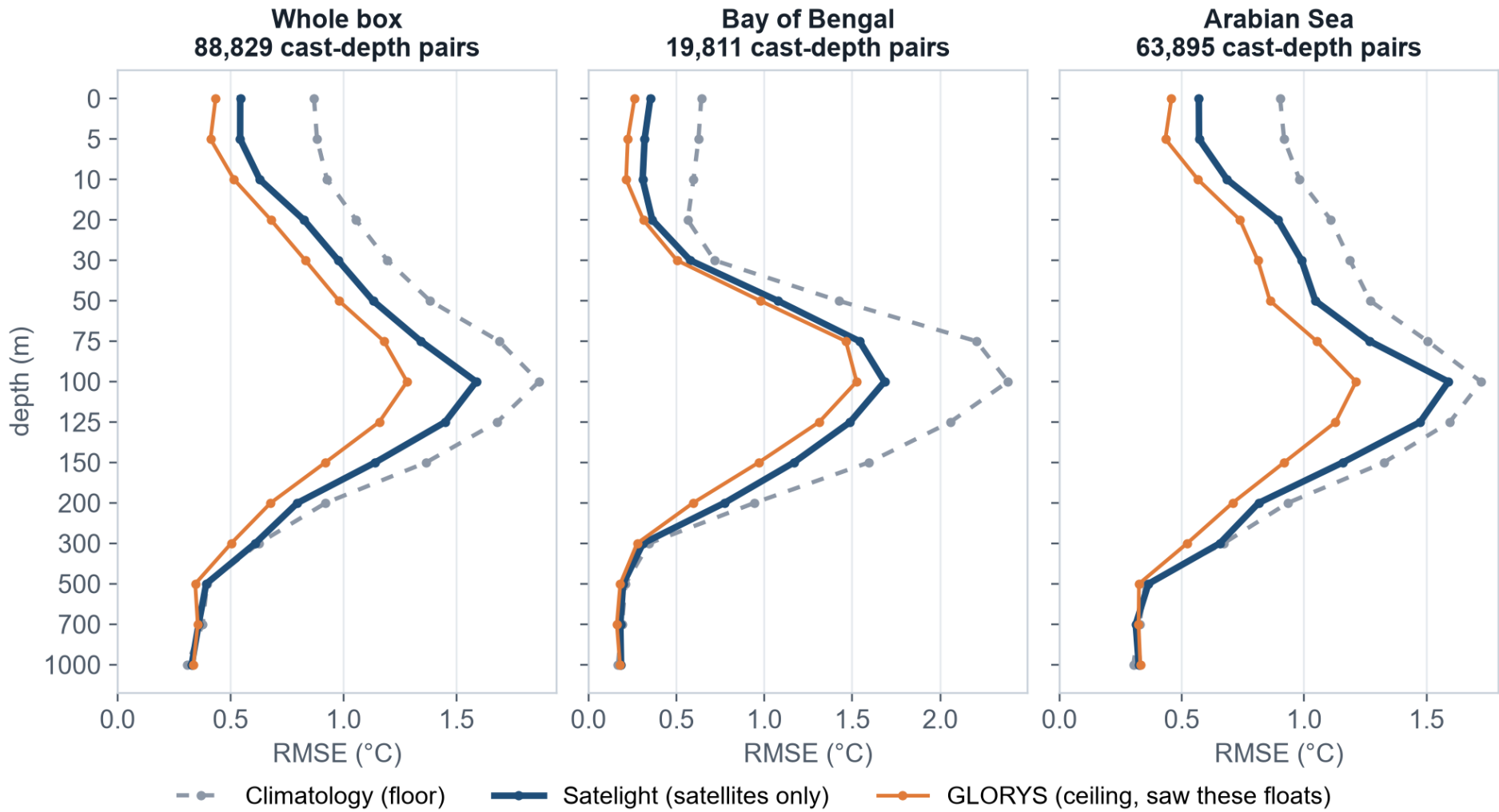
8,500

Argo levels failing QC: counted in the report, never silently dropped

21.0 → 12.9

kJ/cm² RMSE in cyclone heat potential vs Argo: climatology → ours (GLORYS 11.9)

WHAT THE HARNESS FOUND: ERROR AGAINST HELD-OUT ARGO, BY DEPTH



From the surface to 300 m Satelight beats climatology in both basins and stays behind GLORYS, which assimilated these floats. **The thermocline (75–150 m) is the hard part:** errors peak there and run warm.

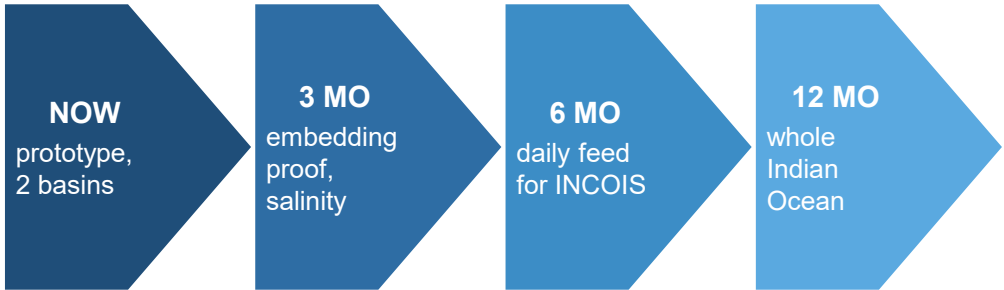
CHALLENGES AND RISKS, AND OUR STRATEGIES

CHALLENGE OR RISK	STRATEGY, ALREADY IN THE BUILD
GLORYS already contains the Argo floats	GLORYS is never an input; it is scored as the ceiling, not a rival.
Few years of satellite salinity	Contiguous time blocks; recipe chosen on the validation years only.
The test years are warmer	An anomaly target; the warm bias at 75–150 m is reported, not hidden.
Sources on different grids and clocks	Block means only, never upsampled; each assumption written into the file.
The embedding proof is half met	unet-abs's embedding beats the raw inputs on mixed-layer depth (R^2 0.351 vs 0.255); a second encoder is next.

SUSTAINABILITY

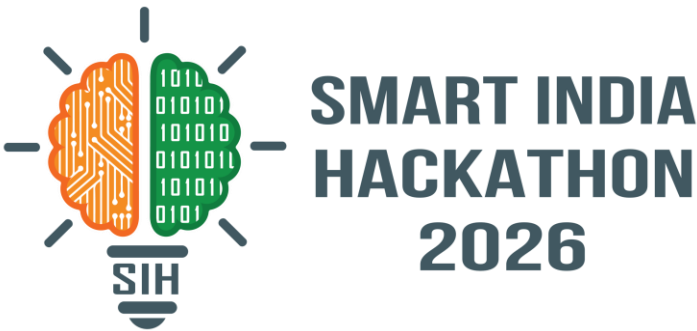
0 Rs running cost; free public data
New source = one fetch function
Every run from a config and a seed
Open source; issues and PRs open

ROADMAP



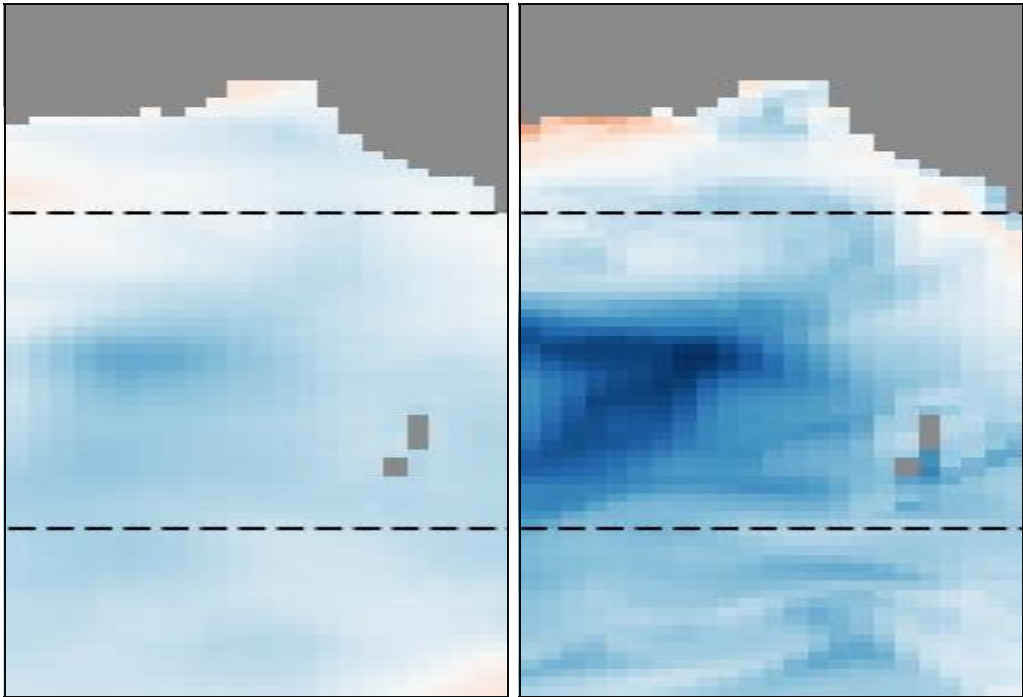


IMPACT AND BENEFITS



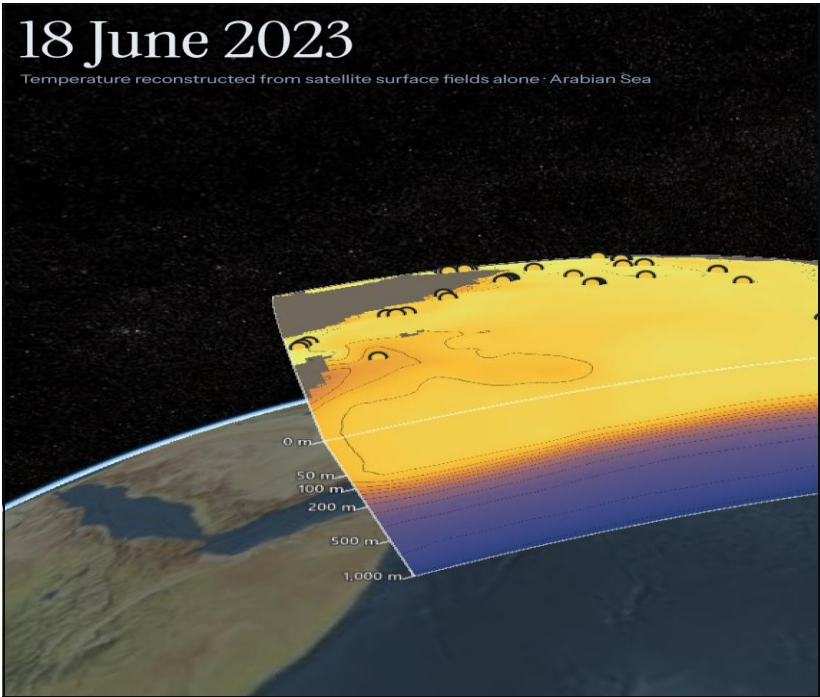
EACH TYPE OF USER AND THE IMPACT ON THEM

INCOIS cyclone forecasters



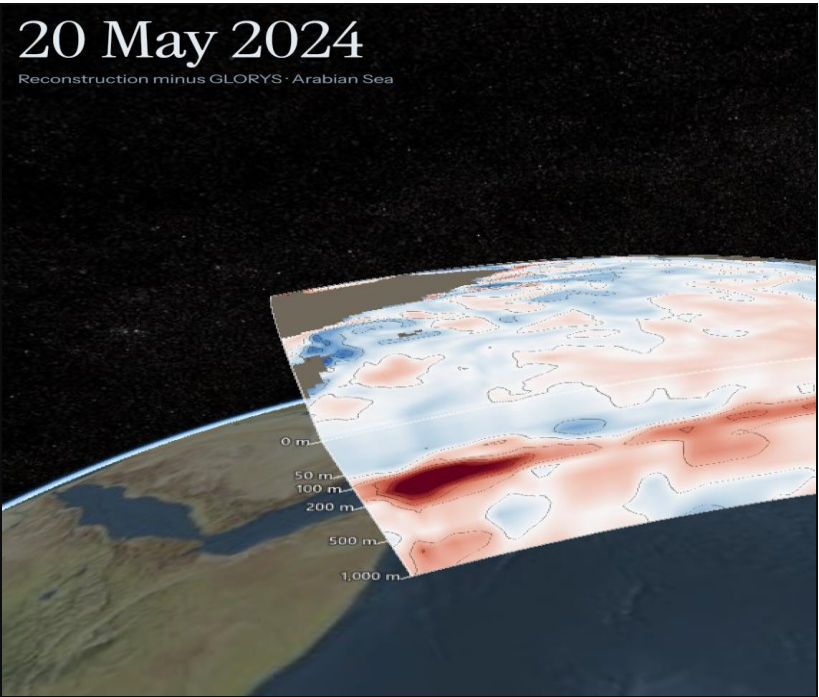
Satelight
Cyclone Mocha, 2023-05-06 to 17: box-mean heat potential 93.6
→ 80.0 kJ/cm², satellites only.

Fisheries



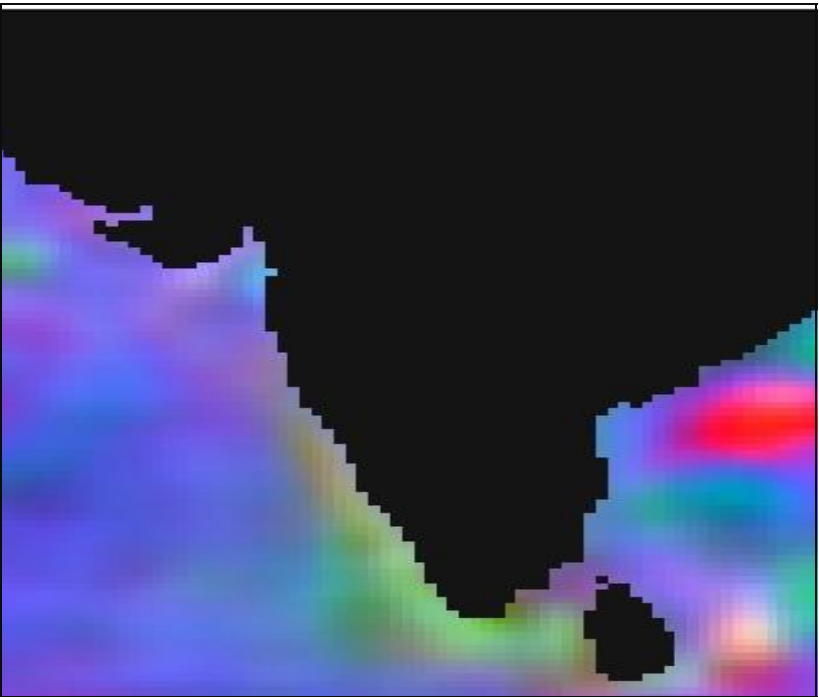
Where the warm layer ends, every day at 0.25°: the thermocline depth that pelagic fish follow, between the floats.

Researchers and modellers



Reconstruction minus GLORYS in 3D, day by day: where satellites can see below the surface and where they cannot.

Public and policymakers



The embedding as colour: each day's ocean summarised by the model, water that looks alike painted alike.

BENEFITS

Scale: one model covers the whole North Indian Ocean box; a new region is a config line and a retrain. Next: salinity at depth, heatwaves.

Social

- Disaster preparedness: a cyclone's fuel, daily, between the floats
- Cyclone wakes seen below the surface from satellites alone
- Open data and code for Indian ocean science and teaching

Economic

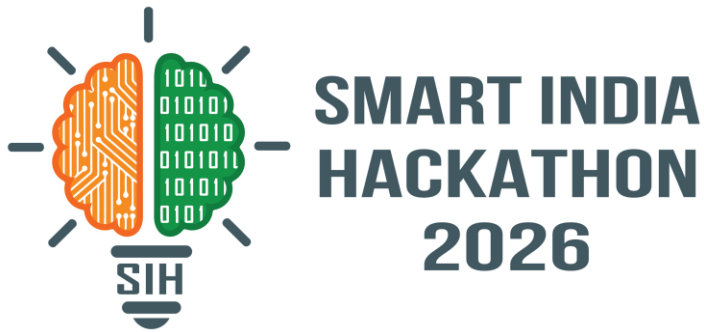
- Uses satellites already in orbit: no new instruments to buy
- One 8 GB GPU; free public data; open source
- Fills the gaps between Argo floats, which are costly to deploy

Environmental

- Upper-ocean heat content every day, for climate monitoring
- Marine heatwaves below the surface, not only at it
- Every score shown beside climatology and GLORYS, so decisions know the limits



RESEARCH AND REFERENCES



Source code, docs and every measured number: github.com/Kukyos/Satelight

DATA, EACH SOURCE PROBED

1

Copernicus Marine GLORYS12 reanalysis, daily. The training target and the ceiling.
data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030

2

OSTIA SST and multi-observation SSS, reprocessed. Two inputs.

data.marine.copernicus.eu

3

DUACS sea level, SLA and ADT. Input; the common grid.
data.marine.copernicus.eu/product/SEALEVEL_GLO_PHY_L4_MY_008_047

4

OSCAR L4 v2.0 surface currents, NASA PO.DAAC. Input.
podaac.jpl.nasa.gov/dataset/OSCAR_L4_OC_FINAL_V2.0

5

CCMP v3.1 10 m winds, Remote Sensing Systems. *Input*.
podaac.jpl.nasa.gov/dataset/CCMP_WINDS_10M6HR_L4_V3.1

6

Argo profiles via Ifremer ERDDAP. The held-out validation.
erddap.ifremer.fr/erddap

7

INCOIS ERDDAP, Argo 10-day analysis. Second reference.
erddap.incois.gov.in/erddap

8

IMD RSMC New Delhi, Mocha and Biparjoy, 2023. Storm dates.
rsmcnewdelhi.imd.gov.in

DATA AND METHODS

9

Wong et al., Argo Quality Control Manual. QC flags, range test.
doi.org/10.13155/33951

10

Ronneberger et al. (2015), U-Net. The convolutional encoder.
doi.org/10.1007/978-3-319-24574-4_28

11

Vaswani et al. (2017), Attention. The hybrid's transformer.
arxiv.org/abs/1706.03762

12

CF Conventions for NetCDF. The output file.
cfconventions.org

13

IOC, SCOR, IAPSO (2010), TEOS-10. Depth, heat capacity.
teos-10.org

14

Leipper & Volgenau (1972), J. Phys. Oceanogr. 2. Heat above 26 °C.
[doi.org/10.1175/1520-0485\(1972\)002<0218:HPOTG>2.0.CO;2](https://doi.org/10.1175/1520-0485(1972)002<0218:HPOTG>2.0.CO;2)

15

Thyng et al. (2016), Oceanography 29(3). cmocean palettes.
doi.org/10.5670/oceanog.2016.66

16

CesiumJS. The globe and renderer.
cesium.com/platform/cesiumjs

WHAT THE SCIENCE CITES

17

Gray (1968), Mon. Weather Rev. 96, 669–700. Cyclones need water above ~26 °C.

18

Price (1981), J. Phys. Oceanogr. 11, 153–175. A cyclone cools the sea.

19

Schott & McCreary (2001), Prog. Oceanogr. 51, 1–123. The monsoon current reverses.

20

Loshchilov & Hutter (2019), Decoupled weight decay. The optimiser.

21

Lellouche et al. (2021), Front. Earth Sci. 9. GLORYS12 and what it assimilates.

22

Argo Program, "How Argo floats work". Floats.
argo.ucsd.edu

23

SIH 2026 problem statement 26066, quoted verbatim.
github.com/Kukyos/Satelight/blob/main/docs/01-problem-statement.md